

LUAT (LUKE) NGUYEN

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CORE COMPETENCIES

- **Site IT Leadership & Engagement:** Directed end-to-end IT delivery as the sole on-site lead for major life sciences research facilities. Proven ability to partner with local scientific leadership to align laboratory operations with global digital strategy.
- **Laboratory Engineering & Computing:** Deep expertise in the technical evaluation, installation, and management of laboratory instrumentation, instrument control systems, and scientific computing hardware (LIMS, ELN, NGS clusters).
- **IT Service Management (ITSM) & Endpoint Operations:** Manage day-to-day lab computing services, endpoint device onboarding, and ad-hoc troubleshooting, ensuring robust and secure connection of instrumentation into the enterprise environment.
- **AI Enablement & Predictive Analytics:** Oversee the strategic deployment of advanced AI/ML solutions, cloud-based collaboration tools, and high-performance computing to augment scientific discovery and process automation.
- **Compliance & Cybersecurity:** Ensure strict data security, regulatory compliance (GxP, data privacy), and audit readiness through zero-trust models, vulnerability management, and fine-grained access controls.

WORK EXPERIENCE

4D Molecular Therapeutics | Emeryville, CA

Senior Infrastructure Engineer/Business Analyst (Contract) | September 2024 – March 2025

Reporting to the Head of IT Infrastructure, Operations, and Security:

- Delivered comprehensive IT Service Management for LabOps, overseeing endpoint device onboarding, ad-hoc troubleshooting, and cyber security services across GxP and non-GxP workflows.
- Managed a team of technical contractors to install, configure, and evaluate cloud-native computing hardware and lab software, ensuring business continuity and rapid resolution of technical issues.
- Spearheaded vulnerability management initiatives to enforce zero-trust security and ensure compliance with regulatory and data security standards for scientific instrumentation.
- Acted as the primary business relationship management liaison, proactively advising leadership on risks and collaborating directly with R&D teams to resolve scientific data pipeline issues.

Merck | South San Francisco, CA

Senior Specialist, Product Analyst, Technical Product Management | January 2022 – March 2024

- Collaborated with R&D scientists and IT executives to initiate, plan, and set strategic direction for an end-to-end digital architecture roadmap aligning with global discovery pipelines.
- Led matrixed, cross-functional global teams in deploying complex AI/ML platforms and advanced lab technologies, ensuring seamless integration with local computational workflows.
- Led the technical product analysis for an enhanced Research Operation Plan, heavily integrating data-driven AI/ML strategies to accelerate research (earned an [INSPIRE Award](#) for innovation).
- Promoted a customer-focused mindset by streamlining IT system architectures, reducing manual operational overhead, and proactively advising leadership on project status, emerging lab technologies, and integration risks.

Senior Specialist, Business/Tech Analysis, Discovery Research IT | October 2017 – December 2021

- Served as the sole on-site Merck IT lead for a new research facility, driving the development and execution of business-aligned IT (Windows/Linux/Clouds) strategies, site policies, and best practices.
- Delivered IT support for site relocations, infrastructure upgrades, decommissioning activities, and data management (compute, storage), managing endpoint device lifecycles and secure infrastructure in compliance with enterprise security standards.
- Partnered with cross-functional IT teams and business stakeholders to assess hardware/software suitability and deliver tailored IT solutions that enhanced user productivity to support business growth.
- Developed SQL queries and dashboards for sample validation, registration, and usage pattern analysis.
- Designed and deployed custom cloud-based applications (Microsoft ecosystem) to centralize instrument data capture and research equipment management.
- Empowered scientific end-users by coordinating training and authoring detailed documentation on system usage, architectural design, and operational best practices.
- [Contributed to the South San Francisco MRL Extended Leadership Team \(XLT\), fostering a unified site culture and strengthening enterprise-wide leadership.](#)
- Received [24 INSPIRE awards \(2017 - 2024\)](#) for contributions to enterprise architecture, cross-functional collaboration, and R&D innovation

Medivation / Pfizer | San Francisco, CA

Senior Business Analyst, Drug Discovery Systems | March 2014 – February 2017

Served as go-to person driving strategic IT partnerships in Research, Informatics, and AI/ML

- Spearheaded requirements gathering and systems integration, bridging the gap between life sciences researchers and agile engineering teams.
- Administered and integrated key LIMS, ELN, and scientific data management systems (CDD Vault, Schrodinger, Spotfire) into R&D projects to support robust drug discovery.
- Served as BA lead for in-house database-driven (agile) application development (recognized for IT Excellence & Innovation).
- Coordinated scientific endpoint device management, ad-hoc IT support, and architectural data storage optimization for a Next Generation Sequencing (Linux cluster) environment.

Evotec | South San Francisco, CA

Informatics Associate, Project Management | May 2011 – February 2013

- Engineered scalable data automation pipelines and high-throughput ETL processes to support complex scientific research and laboratory operations.
- Designed and implemented custom database queries and a web application, significantly improving LIMS data accessibility and visualization for scientific end-users.
- Ensured high-fidelity data management through rigorous QC, optimizing sample inventory and instrument data capture.

VM Discovery, Inc. | Fremont, CA

Computational Research Specialist, Modeling & Informatics | September 2003 – June 2009

- Generated predictive, data-driven insights to advance small molecule therapy research, integrating algorithmic analysis into the early-stage drug discovery process.
- Co-authored patents [US8586619B2](#) and [US61/120,827](#) for first-in-class drug discovery programs.
- Contributed to the successful exit of the TrkA program to Purdue Pharma (up to \$213M) and secured \$2M in NIH grants for the PKC Epsilon program.

Proteomic Systems, Inc. | Sunnyvale, CA

Research Scientist, Membrane Biology | February 02 – February 03

- Designed and conducted fluorescence-based assays to validate and optimize micro-arrayed MembraneChip technology (Patent [US6699719B2](#)).
- Planned and executed complex experiments, analyzing data to enhance MembraneChip performance and inform platform improvements.
- Collaborated with cross-functional teams to resolve technical issues and refine assay protocols.
- Contributed to a peer-reviewed publication by providing key data insights and experimental analysis.

ArQule, Inc. | Menlo Park, CA

Research Scientist, Computational Modeling | February 00 - October 01

- Designed and implemented informatics solutions, leveraging data mining and algorithm development to enhance ADMET (Absorption, Distribution, Metabolism, Elimination) predictive models.
- Developed machine learning-based predictive models for ADMET data, improving model accuracy and decision-making.
- Created molecular descriptors to refine model performance and enhance interpretability.
- Integrated new methods into MOE (Molecular Operating Environment) software, automating complex data analysis and reducing manual errors through streamlined workflows.
- Built interactive data visualization tools and dashboards to support real-time analysis, facilitating data exploration and informed decision-making.

EDUCATION & CERTIFICATIONS

B.S. in Biochemistry/Chemistry | University of California, San Diego

Area of Focus Requirement: Economics & Computer Science

Selected Technical & Scientific Coursework (MOOC):

- **AI & Cloud Integration:** Intro to LLMs, Generative AI, Responsible AI; AWS Fundamentals: Going Cloud-Native, Addressing Security Risk (AWS); Machine Learning with Big Data & Apache Spark.
- **Life Sciences:** Drug Discovery & Development; Python & Command Line Tools for Genomic Data Science; Principles of Biopharmaceutical Manufacturing.